

Appendix C. Runtime and Resource Log

C.1 Development Environment

Item	Details
Development Platform	Kaggle Notebook Environment
Framework	PyTorch
GPU Used	Kaggle GPU T4×2
VRAM	16 GB
Inference Strategy	Sliding-window inference with TTA
Mixed Precision	Enabled where supported

C.2 Approximate Runtime Summary

Activity	Approximate Runtime
Dataset inspection and EDA generation	~20-30 min
Training and validation experiments	~2-2.5 hrs
Threshold calibration and postprocessing sweeps	~20-30 min
Final test inference with TTA	~10-15 min
Qualitative visualization export	~10 min

C.3 Runtime Notes

Most experiments were conducted in a Kaggle notebook environment using a lightweight early-fusion U-Net pipeline. The implementation prioritized faster experimentation cycles and lower GPU usage rather than large multimodal architectures. The training pipeline was executed with an average of 9.5 minutes per epoch.

Sliding-window inference with test-time augmentation increased inference time moderately during final evaluation.